

Introduction to MERN

MERN -

M - MongoDB

E - Express JS

N - Node JS

R - React JS

SD - Frontend - React JS

Backend - Node JS, Express JS

Database - MongoDB

Introduction-

The MERN stack is a widely adopted full stack development framework that simplifies the creation of modern web applications.

It's a full-stack JavaScript stack, meaning we can use JS both on client - or server side leading to consistency and ease of development.

Why learn MERN stack?

- Single language (JavaScript): No need to learn multiple languages for frontend & backend.

- Fast and Scalable - Node JS and Mongo DB handle large data and high traffic efficiently.
- Popular & In demand - Its open source

~~Four~~ Components of MERN

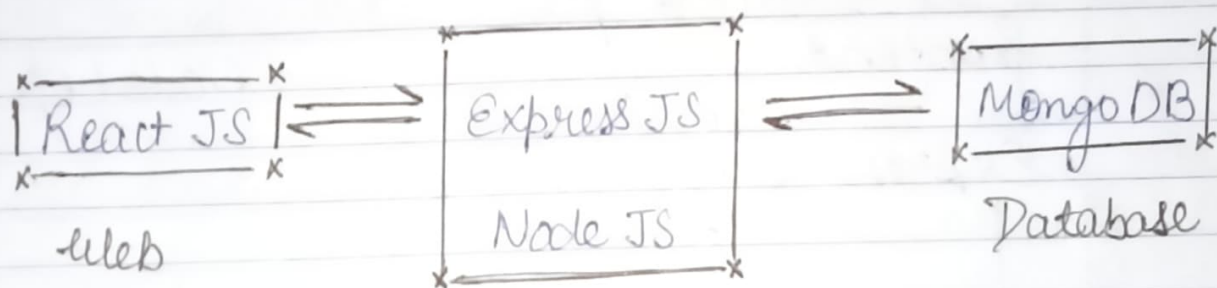
- Mongo DB (Database)
 - A NoSQL database that stores data in JSON like documents instead of tables like traditional databases.
- Express JS (Backend framework)
 - A light weight web framework build on the top of the Node.JS. It simplifies writing backend logic like handling API requests, managing routes, (URL), connecting frontend to Databases.
- React JS (Frontend Library)
 - A JS library used to build user interfaces, especially single page applicⁿ

Its component-based, fast, and helps developers create useable and dynamic UI elements.

• Node JS (Backend Runtime Enviro.)

• This is also JS runtime enviro. that helps lots. you run JS on the ~~server~~ side. Its fast, scalable, and widely used for building backend services and API's

✱ How MERN Stack works? -

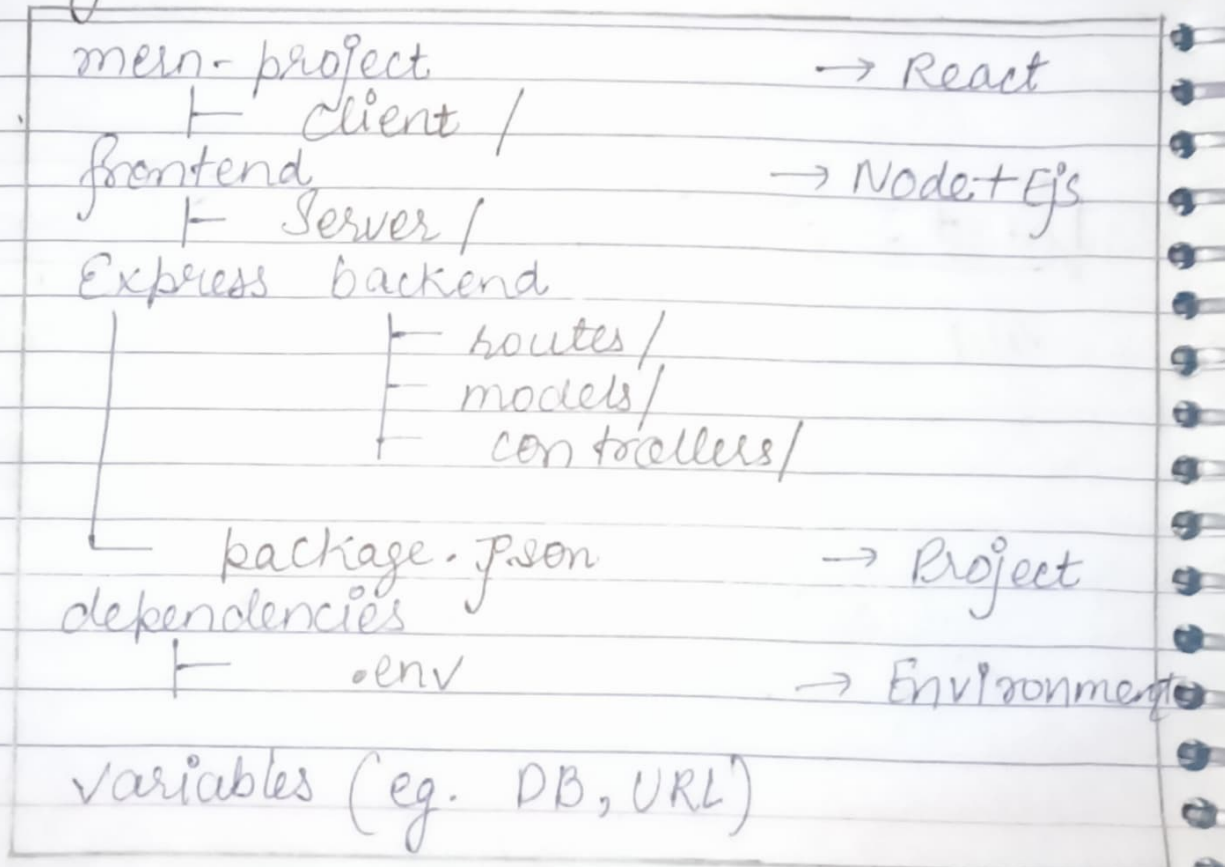


Working of Mern Stack -

↳ Roadmap to be a MERN stack developer

- Learning basics of HTML, CSS & JavaScript.
- Learning React which is a frontend library for building UI.
- Learning Node JS
- Learning Express JS,
- then Learning MongoDB, a NoSQL database to store or retrieve data from databases.

↳ Typical Folder Structure



Advantages of MERN

1. Reuseable components -
On the front-end, React components facilitate re-use and modularity.
2. High Performance
 - React's Virtual DOM optimises UI updates
 - Node's non-blocking I/O allows high performance on server side
3. Active community & ecosystem
All 4 components are open source & have large community support, many packages, tutorials, tools.
4. Flexibility & Modularity.
We can add or replace components, adjust architecture as business needs evolve.